Haptic Feedback for Telerobotic Ultrasound Procedures: Enhancing Safety Through Force Limiting

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2025 14th International Conference on Control, Automation and Information Sciences (ICCAIS) | 979-8-3315-7170-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCAIS67508.2025.11262065

***Abstract*—The development of telerobotic systems for medical diagnostics has made it easier for people in remote areas and areas that don't get enough healthcare to get it. Ultrasound imaging is an important part of non-invasive diagnostics that depends on precise control of probe pressure to make sure the images are clear and the patient is safe. But because telerobotic ultrasound systems don't give direct tactile feedback, they can apply too much or too little contact force, which can make the diagnosis less effective and the patient less comfortable. This paper talks about a telerobotic ultrasound system with haptic feedback that uses a real-time force-limiting algorithm to give the operator more control and lower the risks that come with using too much force. The system has a master-slave structure. On the patient side, there is a UR5 robotic arm with an ultrasound probe and a force sensor. On the operator side, there is a haptic interface that lets the operator feel the force in both directions. Using tissue phantoms to test the system shows that adding haptic feedback to force limitation greatly cuts down on force overshoot and makes tasks more precise. This makes the system safer and better at doing remote diagnostics.**

***Index Terms*—ultrasound images, image processing, organ enhancement, median filter, thresholding**

I. Introduction

The global push for remote medical care has made the need for remote diagnostic tools even greater. Telerobotic ultrasound is a way to reach patients in areas that are hard to get to or do not have a lot of resources [1]. But this method has some problems, the most important of which is that it doesn't give any tactile feedback, which is very important for safe and effective ultrasound procedures. The risk of damaging tissue from too much probe pressure means that haptic technology needs to be used to bring back the sense of touch. This paper describes a force-limiting control system that is built into a telerobotic ultrasound platform. It gives the operator real-time feedback through a haptic device [2].

As healthcare systems move toward digital technology, telerobotics is also moving from experimental setups to real-world use. It is becoming more and more important to understand the nuances of how to physically interact with human tissue, especially when medical procedures are done from a distance. The system must also be safe for, reliable, and effective for diagnosing [3]. Our method solves these problems by using a closed-loop haptic feedback system that can correct mistakes and help users in real time.

Most of the previous research on telerobotic ultrasound has been about how well images are sent and how accurate robotic arms are. Remote ultrasound is possible, as shown by systems like INTERACTS and MELODY. But not many people have talked about the lack of force feedback [4]. Research on haptic interfaces has shown that they can help surgical robots have better control, but they are still not widely used in diagnostic procedures. This paper builds on these results by adding a force-limiting mechanism that works in real time and adapts to the haptic feedback loop. Also, a lot of work has gone into making smart robots that can move like people do. Haptic-enabled robotic surgery platforms like the da Vinci Surgical System have shown that tactile feedback can help make surgery more accurate [5], [6]. To use this principle in diagnostic ultrasound, needs to be change how the user would like to control the force to account for changes in tissue compliance and the needs of the procedure. Ultrasound scanning, on the other hand, requires constant, gentle contact, which shows how important it is to manage dynamic force in a way that takes into account the patient's differences and the scan's context [7].

This research paper talks about a haptic-enabled robotic platform that was made to improve the accuracy of manipulation and the stability of control in environments where tasks are constantly changing. Our platform greatly reduces unintended force transients compared to regular non-haptic systems. These transients can cause performance problems and even damage to hardware or tissue in sensitive applications like surgical robotics or teleoperation. This improvement is made possible by adding high-fidelity tactile feedback mechanisms at the end-effector level. This lets the operator or control algorithm see and respond to contact events in real time. Our method uses continuous, real-time tactile input to allow anticipatory control, unlike passive force sensing systems, which only log interaction forces for later analysis or small reactive changes. This lets the system change on its own in response to changes in contact conditions, surface compliance, and object dynamics. The result is a manipulation framework that is stronger and more responsive, able to keep its accuracy and safety even when the conditions of the interaction are not clear or changeable.

II. System Architecture

The proposed system has a master-slave structure. A Geomagic Touch haptic device on the master side lets the operator move the ultrasound probe and feel the force, while a 6-DOF UR5 robotic arm on the slave side holds the ultrasound probe and has a force sensor (like an ATI Mini45) [8]. In the Communication Layer, a high-speed, low-latency network sends positional and force data, and real-time ultrasound images help the operator [9]. A strain-gauge-based ATI Mini45 six-axis force/torque sensor is mounted between the probe and the robotic flange. It measures forces (F_x, F_y, F_z) and torques (T_x, T_y, T_z) at 1 kHz. This 6-DOF sensing lets you keep an eye on axial pressure and lateral shear very closely, which makes dynamic ultrasound scanning across curved or flexible tissues safer and more stable [10]. In the master-slave setup, the Geomagic Touch sends position and orientation data to the UR5 controller via a low-latency Ethernet link. The UR5 controller then turns this data into joint trajectories. In the opposite direction, the ATI Mini45 sensor sends six-axis force and torque data back to the master with a delay of less than a millisecond to create haptic feedback. This two-way communication makes sure that motion commands and force feedback are sent and received in real time, which is necessary for safe medical procedures.

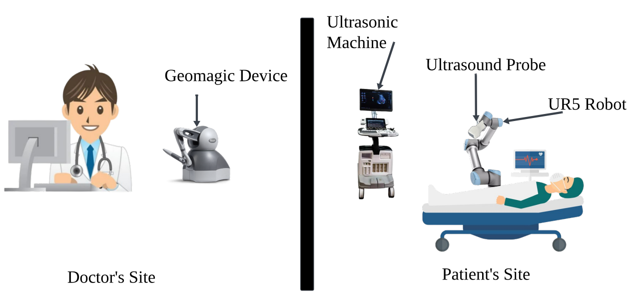

Fig. 1. Telerobotic Ultrasound System

Figure 1 shows how the system is set up [11]. The picture shows how a telerobotic ultrasound system is built so that ultrasound exams can be done from a distance. The Doctor's Site and the Patient's Site are the two main parts of the system. A doctor uses a Geomagic haptic device at the Doctor's Site, which lets them manipulate things in a natural and precise way. The Geomagic device records the doctor's hand movements and force feedback and sends them to the remote site in real time. A UR5 robotic arm at the patient's site copies the movements of the Geomagic device. The ultrasound probe is held and moved over the patient's body by this robotic arm. The ultrasonic machine that the probe is connected to takes live ultrasound pictures and sends them back to the doctor's office so they can see and diagnose them right away. This setup lets the doctor do remote diagnostic tests with a lot of accuracy, making sure that medical imaging happens on time without the need for the doctor to be there in person. In addition to these ultrasound images, a camera installed at the Patient's Site provides a continuous visual feed of the environment, enabling the doctor to monitor the procedure in real time. Together, the ultrasound imaging and the camera view allow the doctor to maintain both diagnostic accuracy and situational awareness, ensuring safe and effective remote examinations. It is very helpful in rural or underserved areas, during pandemics, or when the specialized care is required. The operator's movements are recorded and turned into robotic movements, and the force sensor measures how much pressure is being applied. The haptic interface gets this information back, making sure that the operator feels like they are really touching the patient's body.

The telerobotic system has a UR5 robotic arm that is very precise and can do the same thing over and over again with sub-millimeter accuracy. Each joint has accurate rotary encoders that keep track of where it is, which is very important for safe human interaction. A 3D-printed adapter connects a commercial ultrasound probe to the end effector. This adapter is designed to give the probe both mechanical stability and a little bit of wiggle room so that it can connect with the skin during scans. A multi-axis force/torque sensor from ATI Mini45 is behind the probe and measures axial and lateral forces at 1 kHz [12]. To make sure that all the parts work together safely and well, the cables are properly managed and the vibrations are kept to a minimum. The software is built on a modular, ROS-based architecture that lets it grow, respond in real time, and work with sensors in hardware. The Motion Control Layer uses inverse kinematics to turn the positions of the probes into commands for the joints. This makes sure that the targeting is correct. To reduce sensor noise and process force data to control tactile feedback or change how the probe works, the Force Feedback Layer uses filters [13]. The Safety and Supervision Layer keeps an eye on force and motion at all times to make sure they don't go over set limits. State machines and watchdogs will trigger emergency stops or retractions if conditions become unsafe.The stability of the master-slave loop depends on keeping the communication latency low from a real-time feedback point of view. If there are big delays, the operator's commands might not be carried out right away, which could cause overshoot or loss of fine control. Delayed force feedback can also make things less clear and raise the risk of putting too much pressure on tissue. To fix this, the system uses a high-speed Ethernet link with buffering and synchronization to keep the round-trip delay under 5 ms, which is safe for haptic teleoperation [14]. If there are temporary spikes in latency, the safety layer takes over operator input by holding or retracting the probe to stop unsafe loading. These steps make sure that the system keeps giving reliable real-time feedback, even when the network isn't perfect.

III. Force-Limiting Algorithm

A real-time force-limiting algorithm keeps the contact forces between the ultrasound probe and human tissue within safe physiological limits. This control system uses a PID (Proportional–Integral–Derivative) feedback controller to monitor axial and lateral forces in real time and change the robotic arm's path as needed [15]. The algorithm has two goals. One is to keep the probe in contact with the patient as much as possible with as little discomfort as possible, and the other is to stop too much pressure from building up, which could cause injury or lower image quality. The PID controller changes the end-effector's speed along the contact axis (usually the z-axis in the probe frame) based on how far off the measured force values are from the desired ones. There is also an adaptive gain tuning mechanism to stop oscillations or sudden corrections when moving between tissues that are more or less compliant [16].

A force-limiting algorithm that works in real time and gets feedback controls how the robotic probe interacts with soft tissue [17]. The algorithm is based on a Proportional-Integral-Derivative (PID) control scheme, which constantly checks the sensed force levels against set safety limits and changes the robot's movement as needed. The control loop runs at 1 kHz, which is the same speed as sensor sampling, to make sure that the response time is as short as possible. The PID controller changes the speed and position of the end effector in real time, which allows it to smoothly follow the edges of soft tissue [18]. The gain settings K_p, K_i, K_d are automatically adjusted to keep the system from overshooting or oscillating, especially when switching between tissue types or changing the orientation of the probe [19]. A real-time monitoring routine also constantly checks the size and direction of the forces that are being applied. If a critical threshold is crossed, an emergency stop signal is sent out right away to stop all motion and keep the patient safe.

The control system's force thresholds come from peer-reviewed biomechanical studies and clinical best practices about how much force human soft tissue can handle [20]. For example, research shows that abdominal tissue can usually handle forces up to 10N without causing pain or damage to the tissue [21]. The system is set up to keep applied forces below 4N while it is running, though, to be on the safe side. These limits do not stay the same. They are not fixed; instead, they can be changed based on the patient's anatomy, sensitivity, and the needs of the procedure. The software lets doctors set up force profiles for specific regions, which makes it easier to control force for different clinical situations, like scanning over ribs versus a soft abdomen. The force-limiting module also keeps track of all the force values that were used and events when threshold limits were reached or exceeded to help with future clinical integration [22]. This information can be used to review what happened after the procedure, improve the algorithm, and train. The force-limiting system is governed by a PID controller with force feedback:

$$F_{\text{error}}(t) = F_{\text{ref}} - F_{\text{meas}}(t) \tag{1}$$

The control signal $u(t)$ using a PID controller is:

$$u(t) = K_p \cdot F_{\text{error}}(t) + K_i \cdot \int_0^t F_{\text{error}}(\tau)\, d\tau + K_d \cdot \frac{dF_{\text{error}}}{dt} \tag{2}$$

The variables are defined as follows: F_{ref} represents the reference or safe force threshold; $F_{\text{meas}}(t)$ is the measured contact force at time t; $u(t)$ denotes the control signal sent to the robotic actuator; and K_p, K_i, and K_d are the proportional, integral, and derivative gains of the PID controller, respectively [23].

To prevent overshooting, a saturation function limits the control signal:

$$u_{\text{limited}}(t) = \min\left(\max\left(u(t), u_{\min}\right), u_{\max}\right) \tag{3}$$

To enhance flexibility, an adaptive mechanism recalibrates force thresholds based on tissue stiffness. Stiffness k is estimated using [24]:

$$k = \frac{\Delta F}{\Delta x} \tag{4}$$

Where ΔF is the change in measured force, and Δx is the displacement of the probe.

The telerobotic ultrasound system's UR5 robotic arm has a closed-loop force control architecture, which is shown in Figure 2. The system keeps the contact force at the level set by the remote doctor by comparing it to the actual force measured by a sensor. A PID controller processes the error and sends velocity commands to the robot so that it can move the probe [25]. A force sensor on the probe monitor the contact force all the time, and a safety layer makes sure it stays within set limits and takes action if it goes over them. This feedback loop in real time makes scanning safe and accurate [26]. In the motion command output stage of the control loop, the PID controller regulates probe velocity along the contact axis by adjusting the end-effector speed based on the force error. The response of the UR5 depends on the chosen gains: higher proportional terms improve responsiveness but risk overshoot, integral action helps reduce steady-state error yet can cause oscillations, and derivative action provides damping to suppress sudden force spikes [27]. To achieve safe and stable behavior, the gains were tuned experimentally toward a critically damped response, minimizing overshoot while maintaining accuracy. In practice, lower gains are used during initial probe–skin contact

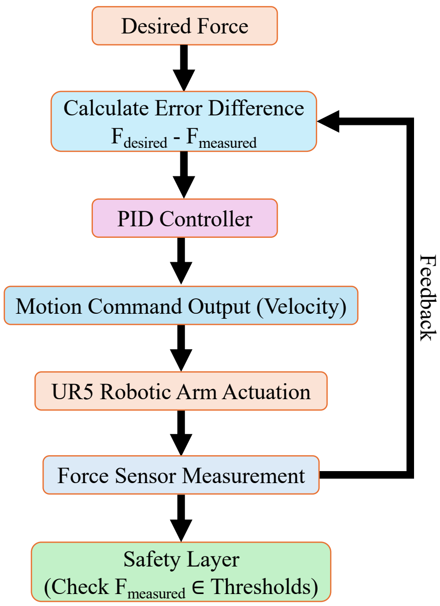

Fig. 2. Force Control Loop for UR5-Based Telerobotic Ultrasound System

to avoid abrupt loading, moderate gains sustain steady contact during scanning, and derivative-dominant adjustments prevent excessive force if thresholds are approached. These schemes together ensure reliable contact maintenance and patient safety across varying tissue conditions.

IV. Results and Discussion

Figures 3 and 4 show ultrasound pictures of the right (RK) and left kidneys (LK) that were taken from a distance using the proposed telerobotic ultrasound system [28]. A Geomagic haptic interface let a doctor control the UR5 robotic arm. The RK image in Figure 3 clearly shows the renal cortex, medulla, and echogenic sinus, which means that the probe was in the right place and making good contact. Figure 4 also shows the LK, with the parenchyma and central sinus fat clearly visible. The hypoechoic and hyperechoic areas show that the system can take images that are useful for diagnosis on both sides. Stable probe movement, the right amount of force, and real-time control—made possible by a PID-based force feedback loop and safety threshold—guaranteed safe, high-quality imaging. These results show that the system could be used for ultrasound diagnostics in places where resources are limited or far away [29].

Figure 5 shows the force that the ultrasound probe measured on the patient's skin over the course of about 340 seconds during a telerobotic ultrasound procedure that didn't have any rules about how much force to use. The vertical axis shows the force in Newtons (N), and the horizontal axis shows the time in seconds (s). The graph shows that the force profile changes a lot, with sudden peaks and valleys during the scanning session. At certain points, the force applied suddenly rises sharply, reaching levels as high as 15–16 N, which is far above the best

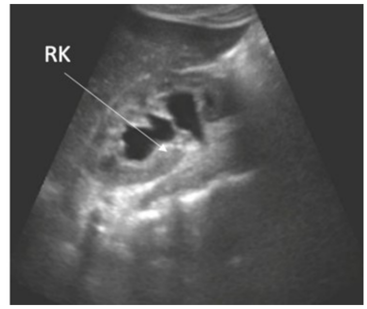

Fig. 3. Ultrasound image showing the Right Kidney (RK)

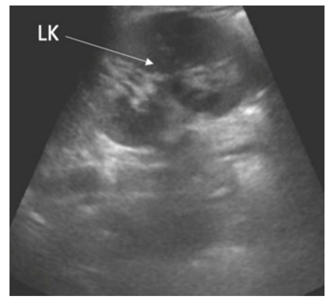

Fig. 4. Ultrasound image showing the Left Kidney (LK)

range for diagnostic ultrasound imaging. The remote operator probably caused these spikes by accidentally going too far or making corrections too late. This could be because the control system doesn't have real-time haptic feedback or has variable latency. On the other hand, there are times when the force drops to almost 0 N, which means that the probe isn't always in contact with the skin. This could lower the quality of the images and the accuracy of the diagnosis [30]. The fact that the force curve is not smooth shows how hard it is to keep a steady and safe contact pressure when there is no closed-loop control or force thresholding. These kinds of problems can not only lower the quality of the ultrasound images, but they can also make patients uncomfortable or even put them in danger, especially during long exams or in sensitive areas of the body. Also, the way force is applied changes depending on how flexible the tissue is, how the anatomy curves, and how the probe is pointed. When these factors are combined with the fact that there is no automatic regulation mechanism [31], the resulting force profile is unpredictable and not uniform. This observation makes it clear how important it is to use a real-time force feedback system or an autonomous control layer, like a PID controller with set safety limits, to make sure that the force applied stays within a safe and desirable range during the whole procedure [32].

Figure 6 shows how the force changed during a telerobotic ultrasound session where the contact force was actively controlled to stay at a steady 4N. The force trajectory stays very close to 3.5N and 4.5N, which shows how well the closed-loop control keeps the probe pressure almost constant over time. A real-time force feedback system that changed the position

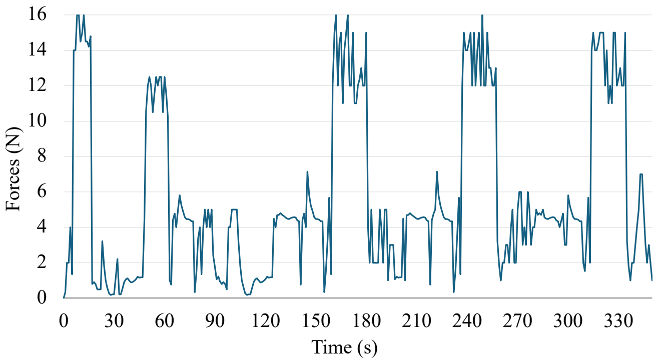

Fig. 5. Unregulated Force Profile During Telerobotic Ultrasound

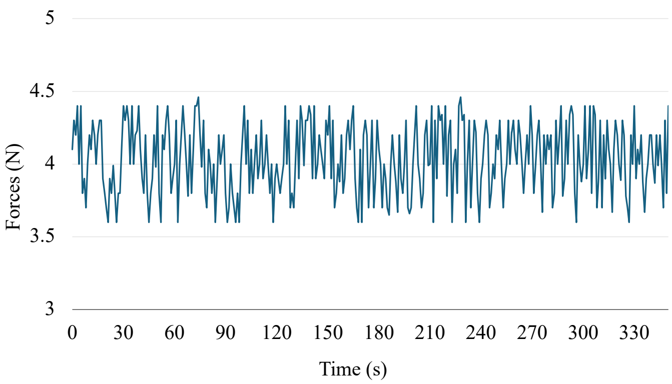

Fig. 6. Regulated Force Application During Telerobotic Ultrasound

of the end-effector in response to small changes in contact force made this regulation possible. It is very important to keep a consistent force with such precision, especially when scanning from a distance or on its own, where patient safety and diagnostic quality must be guaranteed without direct physical intervention. The small range of fluctuations shows that the probe and the patient's body are interacting in a stable way, which helps keep the image quality consistent and lowers the chance of discomfort or tissue stress. Also, the force-controlled environment makes sure that the scanning process is predictable and repeatable, which is important in clinical settings where standardization across multiple sessions or operators is necessary. This fixed-force implementation shows that the system can scan gently, steadily, and with control, which are all important for making telerobotic ultrasound procedures safer and more reliable. Furthermore, to ensure patient safety in case of any unexpected failure, an emergency stop switch is provided in the patient's hand. If the procedure becomes uncomfortable, the patient can press this switch, which immediately resets the robotic system and halts the scanning process.

Figure 7 shows what patients thought about the telerobotic ultrasound procedure when the force was controlled. It focuses on three things; where F1: Overall comfort, F2: Controlled and safe pressure from the robotic probe, and F3: No pain or discomfort. Most patients said they felt better, especially for F1 and F2, which shows that fixed contact force made them feel much safer and more comfortable. The results for F3 also showed very few reports of pain, which supports the idea that the system is easy to use for patients. There weren't many moderate responses, which shows that there were only small differences in sensitivity. There also weren't many negative responses. In general, the regulated-force approach made things much more comfortable, safe, and tolerant. Figure 8 shows what doctors thought of the haptic-feedback system with force-limiting control. They rated it on three things: F1: how well it helped them feel the force, F2: how well it kept them from applying too much pressure, and F3: how much it helped keep patients safe. Most people agreed on all points, with the most support for F2, which shows how well the force-limiting system worked. F1 and F3 also got a lot of positive feedback, which means that patients are safer and more aware of their surroundings. These results show that haptic and force-limiting feedback can help improve the performance and results of remote ultrasound. The proposed system uses commercially available parts like the UR5 robotic arm, the ATI Mini45 sensor, and the Geomagic Touch haptic device to save money and make it easier to maintain. The modular design of this system makes it cheaper overall, even though the initial cost is higher than that of standard ultrasound systems. This is because it allows for incremental upgrades and the use of existing clinical ultrasound machines. The system can be changed into portable or semi-mobile setups for large-scale use, which makes it possible to set up in rural clinics, ambulances, or telemedicine centers.

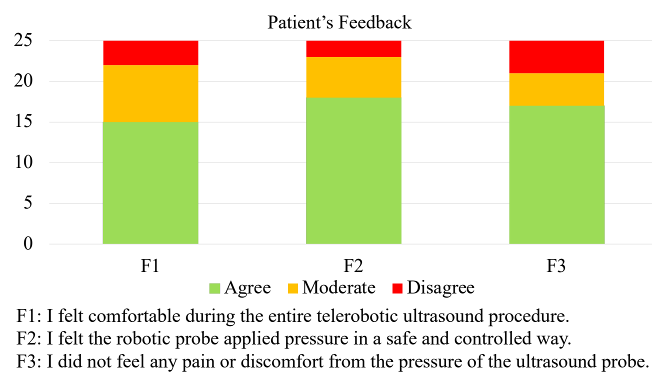

Fig. 7. Patient Feedback on Telerobotic Ultrasound Experience

V. Conclusion

This study shows that adding haptic feedback to force-limiting control makes telerobotic ultrasound procedures much safer, more comfortable, and more reliable. The system lowers the risk of patient discomfort or tissue damage while ensuring stable image acquisition by keeping contact forces consistent and safe. The results show that these systems can be just as accurate as regular ultrasound exams in remote areas,

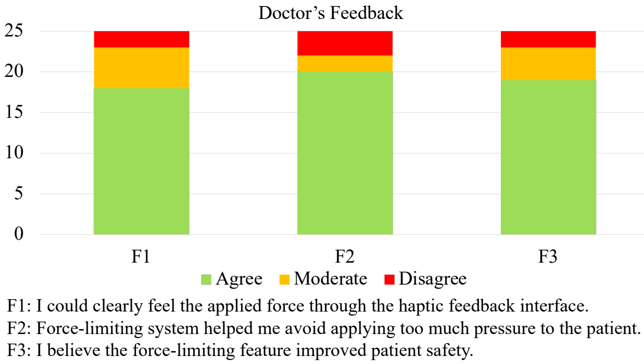

Fig. 8. Doctor Feedback on Telerobotic Ultrasound Experience

making them a good choice for telemedicine. In the future, the focus will be on clinical deployment, where adaptive control algorithms can be improved to change forces in real time based on how the tissue interacts with them. Adding AI-based tissue classification will make force thresholds even more personal and improve image quality. Working with healthcare professionals will help make design changes that make sure the product is usable and accepted in clinical settings. Long-term goals include making portable telerobotic ultrasound units for ambulances and rural care. These units will be able to connect to the cloud and use AI to help with diagnostics, so that experts can see images in places that do not have a lot of resources.

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